One-Page Individual Reflection

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During this assignment, I worked on the design and development of a Hotel Accommodation and Reservation Management System using C++. The main design decision was to model hotel entities such as guests, rooms, reservations, and services as separate classes. I learned that inheritance and polymorphism are especially useful when several room or service categories share common operations but have different implementations.

One of the major challenges was deciding how the classes should interact without making the program difficult to maintain. Handling room availability, reservation cancellation, service charges, and bill calculation required careful validation and testing. I also learned the practical difference between an abstract base class and a derived class, and how virtual functions allow a base-class pointer to invoke the correct derived-class implementation at runtime.

The assignment improved my understanding of encapsulation, inheritance, abstraction, pointers, virtual functions, runtime polymorphism, constructors, STL containers, and modular program design. Testing different situations such as valid reservations, unavailable rooms, invalid reservation IDs, service selection, and cancellation helped me understand why validation is important in real software systems.

The project is also connected with SDG 8 and SDG 9 because software-based hotel management can improve operational efficiency and demonstrates the use of digital technology in the hospitality sector. Overall, the assignment helped me connect C++ classroom concepts with a practical application and improved my problem-solving, coding, testing, and documentation skills. It contributed to the mapped course outcomes by giving me practical experience in designing an object-oriented solution, applying reusable programming concepts, and validating the behavior of a complete application.